

Reactor Design for Production of Maleic Anhydride

Arwah Faisal

Chemical Reaction Engineering (CHE 321)

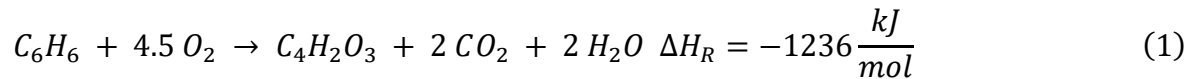
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OBJECTIVE:

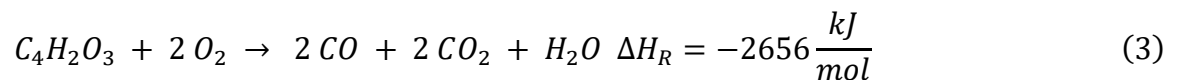
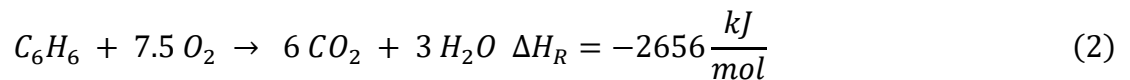
Design a reactor to produce Maleic anhydride from benzene.

REACTION DETAILS:

Maleic anhydride is an industrially important commodity chemical used primarily in the production of polyester resins. An old commercial process to produce maleic anhydride employs a vanadium-phosphorous-oxide catalyst (VPO) to selectively oxidize butane:



However, side reactions to produce CO and CO₂ are also important:



REACTION KINETICS:

Reaction rates for these three reactions are given by:

$$r_1 = \frac{k_1 P_1^{\alpha_1}}{1 + K_2 P_2^2}$$

$$r_2 = k_2 P_1^{\alpha_2}$$

$$r_3 = \frac{k_3 P_2}{(1 + K_2 P_2)^2}$$

where, P₁ is the partial pressure of benzene and P₂ is the partial pressure of maleic anhydride. The following kinetic parameters are suggested (rate constants at 400 °C):

$$k_1 = 1 \times 10^{-6} \text{ kmol/kg s atm}^{0.35}$$

$$k_2 = 1.5 \times 10^{-7} \text{ kmol/kg s atm}^{0.35}$$

$$k_3 = 2.9 \times 10^{-6} \text{ kmol/kg s atm}$$

$$E_1 = E_2 = 83100 \text{ J/mol}$$

$$E_3 = 155000 \text{ J/mol}$$

$$\alpha_1 = \alpha_2 = 0.35$$

$$K_2 = 310 \text{ atm}^{-1}$$

REACTOR CONFIGURATION:

The reactor to be designed is a packed bed reactor with multiple reaction tubes with external cooling. External co-current cooling is to be provided by a mixed salt flowing at 2000 Kg/h. The tube diameter is to be 25 mm OD to maximize heat transfer (Length = 12 ft and thickness = 1/8"). Heat transfer is a significant issue in this reactor due to the large exothermicity of the oxidation reactions.

$$C_p (\text{mixed salt}) = A + BT + CT^2 + DT^3 \text{ (kJ/kmol/K)}$$

MW = 28 kg/kmol

After plugging in the given values for A, B, C, and D, the above equation becomes the following:

$$C_p (\text{mixed salt}) = 50 + 2 \times 10^{-3} T + 1 \times 10^{-5} T^2 - 4 \times 10^{-9} T^3 \text{ (kJ/kmol/K)}$$

DESIGN BASIS:

The feed (assume physical properties of air) will consist of 15 % benzene in a mixture of oxygen and nitrogen (1:1). Total inlet pressure is 2 atm, and the total molar flow rate is 0.1 kmol/hr (per tube). Pressure drop should be considered using the Ergun equation. The following information may be helpful for the design:

Overall heat transfer coefficient (based on outside area) = 120 W/m² K

Catalyst diameter = 2 mm

Void fraction (ϵ) = 0.45

$\rho_B = 3.2 \times 10^3 \text{ kg/m}^3$

Changing the catalyst diameter by a factor of 10 did not affect reactor performance.

DELIVERABLES

You are to consider your design in several parts, and satisfy the following requirements:

1. Model the maleic anhydride reactor assuming an inlet reactant temperature of 300 °C and inlet salt temperature of 150 °C, calculating the conversion of benzene. Plot the pressure, reactor-temperature, salt-temperature and molar flow rates of all the components as functions of reactor length.
2. What will be the analysis for part 1 with counter-current heat exchanger?
3. Plot the following:
 - a. The Pressure Profile for different particle sizes (1, 2, 3, and 4 mm)
 - b. Inlet salt temperature (50, 100, 150, 200, 250, 300 and 350 °C) as a function of the reactor exit temperature and final conversion (for 2 mm particle size).
4. As an engineer, what changes to the design and operating parameters will you suggest to increase the conversion to 85-95%?
5. What safety concerns are there for this reaction? How could you design your reactor to account for these concerns? For example, you might consider the effects of varying the salt inlet temperature on the resulting temperature profiles.

SOLUTION:

Mole Balances: Each species present in the PBR must be considered.

General PBR Mole Balance Equation:

$$\frac{dF_j}{dw} = \sum_i^n r_{ij}$$

All the differential equations must be in terms of dz, in order to convert dw to dz the following equation is used:

$$dw = A_c \rho_B dz$$

$$A_c \rho_B = \frac{\pi D_i^2}{4} \rho_B$$

The general mole balance then becomes:

$$\frac{dF_j}{dz} = A_c \rho_B \sum_i^n r_{ij}$$

The mole balances for each species is the following:

$$\frac{dF_{C_6H_6}}{dz} = (-r'_1 - r'_2)A_c\rho_B \quad (1)$$

$$\frac{dF_{C_4H_2O_3}}{dz} = (r'_1 - r'_3)A_c\rho_B \quad (2)$$

$$\frac{dF_{O_2}}{dz} = (-4.5r'_1 - 7.5r'_2 - 2r'_3)A_c\rho_B \quad (3)$$

$$\frac{dF_{CO_2}}{dz} = (2r'_1 + 6r'_2 + 2r'_3)A_c\rho_B \quad (4)$$

$$\frac{dF_{H_2O}}{dz} = (2r'_1 + 3r'_2 + r'_3)A_c\rho_B \quad (5)$$

The two following equations are intermediates. Air is present; therefore, Nitrogen needs to be considered in the mole balances:

$$\frac{dF_{CO}}{dz} = 0 = 2r'_3A_c\rho_B \quad (6)$$

$$\frac{dF_{N_2}}{dz} = 0 \quad (7)$$

The rate constants must be converted to SI units:

$$k_1 = 1 \times 10^{-6} \frac{\text{kmol}}{\text{kg s atm}^{0.35}} \frac{1 \text{ atm}^{0.35}}{\left(101325 \frac{\text{N}}{\text{m}^2}\right)^{0.35}} \frac{1000 \text{ mol}}{1 \text{ kmol}}$$

$$k_1 = 1.77 \times 10^{-5} \frac{\text{mol}}{\text{kg s}} \left(\frac{\text{m}^2}{\text{N}}\right)^{0.35}$$

$$k_2 = 1.5 \times 10^{-7} \frac{\text{kmol}}{\text{kg s atm}^{0.35}} \frac{1 \text{ atm}^{0.35}}{\left(101325 \frac{\text{N}}{\text{m}^2}\right)^{0.35}} \frac{1000 \text{ mol}}{1 \text{ kmol}}$$

$$k_2 = 2.66 \times 10^{-6} \frac{\text{mol}}{\text{kg s}} \left(\frac{\text{m}^2}{\text{N}}\right)^{0.35}$$

$$k_3 = 2.9 \times 10^{-6} \frac{\text{kmol}}{\text{kg s atm}} \frac{1 \text{ atm}}{101325 \frac{\text{N}}{\text{m}^2}} \frac{1000 \text{ mol}}{1 \text{ kmol}}$$

$$k_3 = 2.86 \times 10^{-8} \frac{\text{mol m}^2}{\text{kg s N}}$$

The reaction rates above can be plugged into each of these equations; however, the rate constants must be determined for any temperature. This is done by using the Arrhenius Equation:

$$k(T) = k(T_0) \exp \left[\frac{E}{R} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right]$$

This equation will be applied to each k:

$$k_1(T) = 1.77 \times 10^{-5} \frac{\text{mol}}{\text{kg s}} \left(\frac{\text{m}^2}{\text{N}} \right)^{0.35} \exp \left[\frac{83100 \frac{\text{J}}{\text{mol}}}{8.314 \frac{\text{J}}{\text{mol K}}} \left(\frac{1}{673.15} - \frac{1}{T} \right) \right]$$

$$k_2(T) = 2.66 \times 10^{-6} \frac{\text{mol}}{\text{kg s}} \left(\frac{\text{m}^2}{\text{N}} \right)^{0.35} \exp \left[\frac{83100 \frac{\text{J}}{\text{mol}}}{8.314 \frac{\text{J}}{\text{mol K}}} \left(\frac{1}{673.15} - \frac{1}{T} \right) \right]$$

$$k_3(T) = 2.86 \times 10^{-8} \frac{\text{mol m}^2}{\text{kg s N}} \exp \left[\frac{155000 \frac{\text{J}}{\text{mol}}}{8.314 \frac{\text{J}}{\text{mol K}}} \left(\frac{1}{673.15} - \frac{1}{T} \right) \right]$$

Ergun Equation: This is used to consider pressure drop in the PBR.

$$\frac{dP}{dz} = - \frac{G}{\rho g_c d_p} \left[\frac{1 - \varepsilon}{\varepsilon^3} \right] \left[\frac{150(1 - \varepsilon)\mu}{d_p} + 1.75G \right]$$

In this equation, everything is known except ρ (gas density), u (superficial velocity), and μ (gas viscosity). These will be calculated, and the above equation will be manipulated.

Viscosity can be determined with the following equation. This allows for the calculation of viscosity as a function of temperature:

$$\mu(T) = 1.7894 \times 10^{-5} \left(\frac{T}{273.15} \right)^{1.5} \left(\frac{273.15 + 110}{T + 110} \right) \frac{\text{kg}}{\text{m s}}$$

The following relationship is valid for a PBR:

$$\dot{m}_o = \dot{m}$$

$$\rho_o v_o = \rho v$$

Using the following equation, the density can be determined:

$$\frac{v}{v_o} = \frac{P_o T}{P T_o} \frac{F_T}{F_{To}}$$

$$\rho = \rho_o \frac{P T_o F_{To}}{P_o T F_T}$$

The ideal gas law can be used to determine ρ_o :

$$PV = nRT$$

$$PV = \frac{m}{MW} RT \rightarrow PMW = \frac{m}{V} RT$$

$$\rho_o = \frac{P_o MW}{RT_o}$$

$$\rho_o = \frac{(2 \text{ atm}) \left(\frac{28.96 \text{ g}}{\text{mol}} \right)}{\left(0.0821 \frac{\text{atm L}}{\text{mol K}} \right) (573.15 \text{ K})} \frac{1000 \text{ L}}{1 \text{ m}^3} \frac{1 \text{ kg}}{1000 \text{ g}}$$

$$\rho_o = 1.2309 \frac{kg}{m^3}$$

Now the last unknown term in the Ergun equation must be calculated.

$$G = \rho u$$

$$u = \frac{v_o}{A_c} = \frac{v_o}{\frac{\pi}{4} D_i^2}$$

The volumetric flow rate at the inlet can be determined using the ideal gas law:

$$P_o v_o = F_{T_o} R T_o$$

$$v_o = \frac{F_{T_o} R T_o}{P_o}$$

$$v_o = \frac{(0.02778 \frac{mol}{s}) (8.314 \frac{J}{mol K}) (573.15 K)}{2 atm} \frac{1 atm}{101325 N/m^2} \frac{N m}{J}$$

$$v_o = 6.532 \times 10^{-4} \frac{m^3}{s}$$

Inner Diameter of PBR:

$$D_i = D_o - 2x$$

Where x is thickness and D_i and D_o are inner and outer diameters of the PBR, respectively.

$$D_i = 25 \times 10^{-3} m - 2 \left(\frac{1}{8} in \frac{2.54 cm}{1 in} \frac{1 m}{100 cm} \right)$$

$$D_i = 0.01865 m$$

$$u = \frac{6.532 \times 10^{-4} \frac{m^3}{s}}{\frac{\pi}{4} (0.01865 m)^2}$$

$$u = 2.391 \frac{m}{s}$$

Now the Ergun equation can be manipulated:

$$\frac{dP}{dz} = - \frac{G}{\rho_o g_c d_p} \left[\frac{1 - \epsilon}{\epsilon^3} \right] \left[\frac{150(1 - \epsilon)\mu}{d_p} + 1.75G \right] \frac{P_o}{P} \frac{T}{T_o} \frac{F_T}{F_{T_o}} \quad (8)$$

Energy Balances: There will be two energy balances for the PBR, one for the reactor temperature and one for the coolant temperature.

$$\frac{dT}{dw} = \frac{\sum r'_i \Delta H_{RXij} - \frac{Ua(T - T_a)}{\rho_B}}{\sum F_i c_{pi}}$$

In terms of dz the equation becomes:

$$\frac{dT}{dz} = \left[\frac{\sum r'_i \Delta H_{RXij} - \frac{Ua(T - T_a)}{\rho_B}}{\sum F_i c_{pi}} \right] A_c \rho_B \quad (9)$$

The last differential equation will be for the co-current coolant balance:

$$\frac{dT_a}{dw} = \frac{Ua(T - T_a)}{\rho_B \dot{m}_c c_{pc}}$$

In terms of dz the above equation becomes:

$$\frac{dT_a}{dz} = \frac{Ua(T - T_a)A_c}{\dot{m}_c c_{pc}} \quad (10)$$

The inlet contains 15% benzene, the initial molar flow rate of benzene can be calculated:

$$F_{C_6H_6_0} = 0.15 F_{T_0}$$

The total inlet molar flow rate is:

$$F_{T_0} = 0.02778 \frac{mol}{s}$$

$$F_{C_6H_6_0} = 0.15 \left(0.02778 \frac{mol}{s} \right)$$

$$F_{C_6H_6_0} = 0.0041667 \frac{mol}{s}$$

Air has a 1:1 mixture of oxygen and nitrogen, therefore, the inlet molar flow rates will be equal:

$$F_{O_{2_0}} = F_{N_{2_0}} = F_{T_0} \frac{(1 - .15)}{2}$$

$$F_{O_{2_0}} = F_{N_{2_0}} = 0.0118 \frac{mol}{s}$$

Dalton's Law of Partial Pressure:

$$P_A = \frac{F_A}{F_{T_0}} P$$

Benzene:

$$P_{C_6H_6} = \frac{F_{C_6H_6}}{F_{T_0}} P$$

Maleic anhydride:

$$P_{C_4H_2O_3} = \frac{F_{C_4H_2O_3}}{F_{T_0}} P$$

At this point all necessary calculations have been made. The important equations, variables, and unit conversions have been recorded. All this information can be plugged into MATLAB to solve the 10 ODEs.

Part 1: Conversion of Benzene and Plots

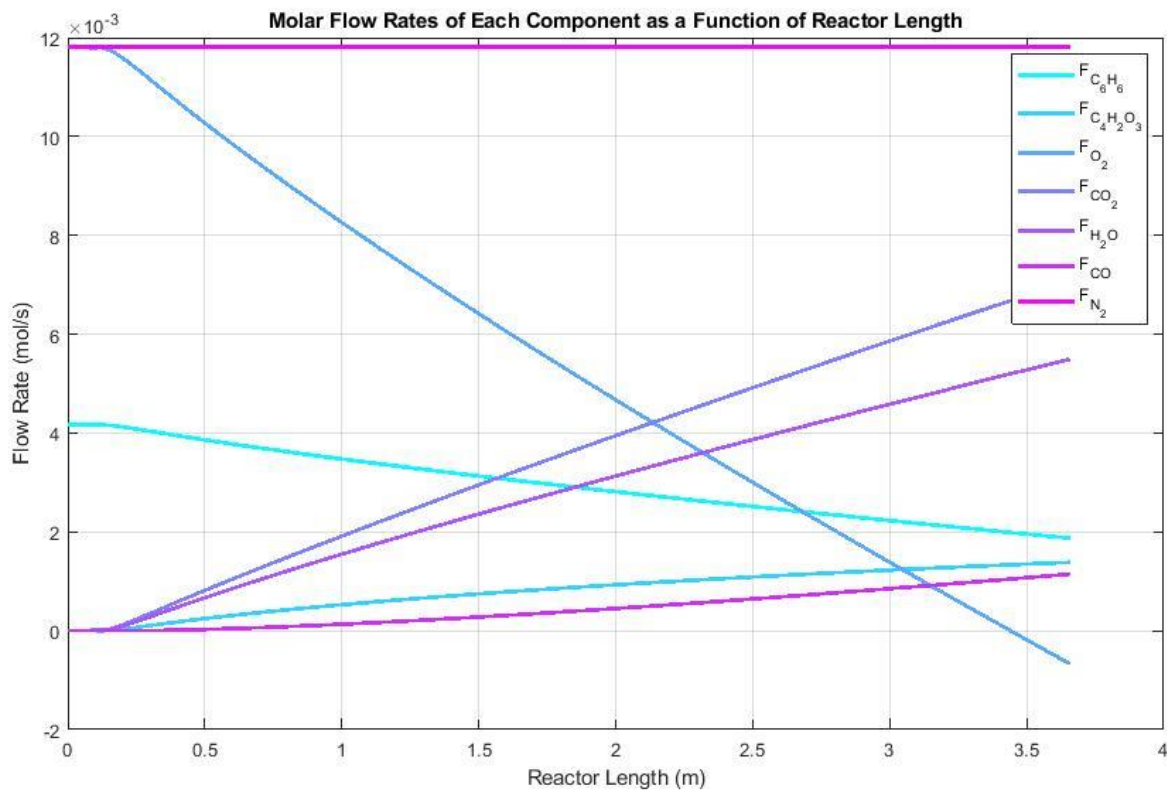
All the above calculations, equations, and variables were plugged into MATLAB to solve the 10 Ordinary Differential Equations simultaneously and graph the results. Each of these equations are labeled accordingly. The MATLAB code is attached at the bottom of this report.

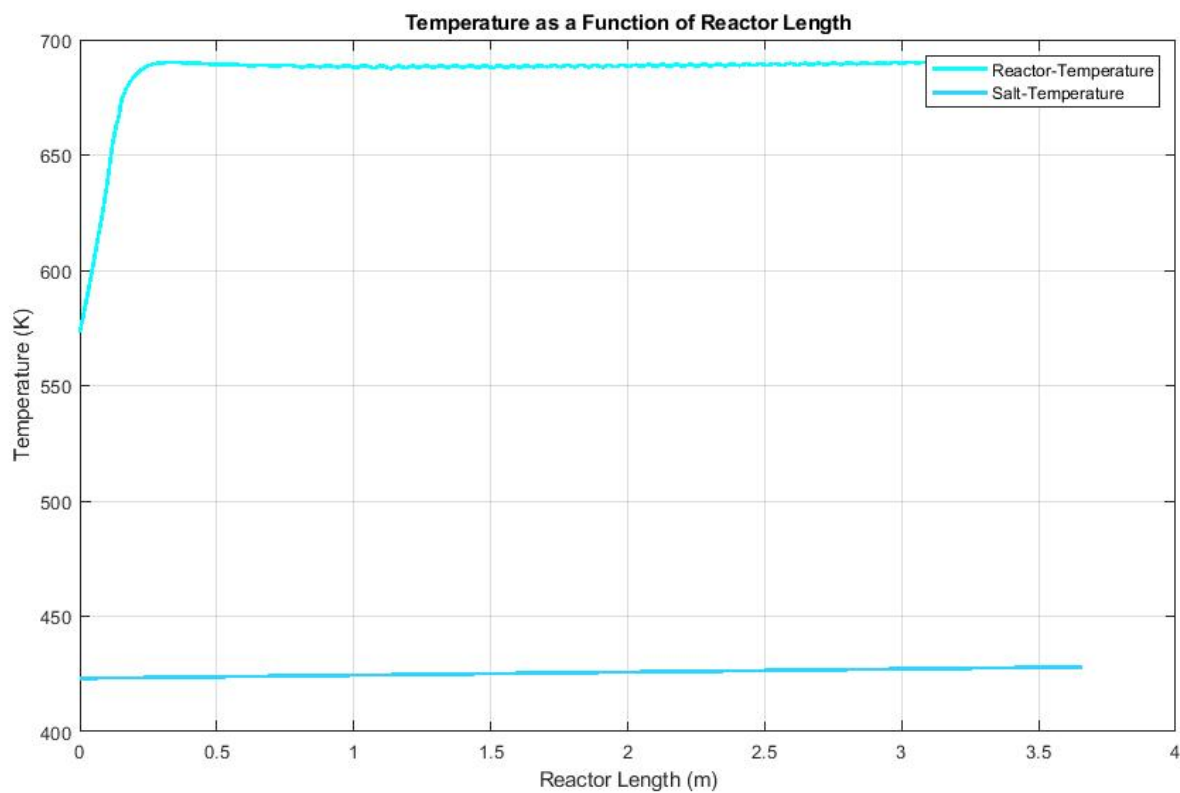
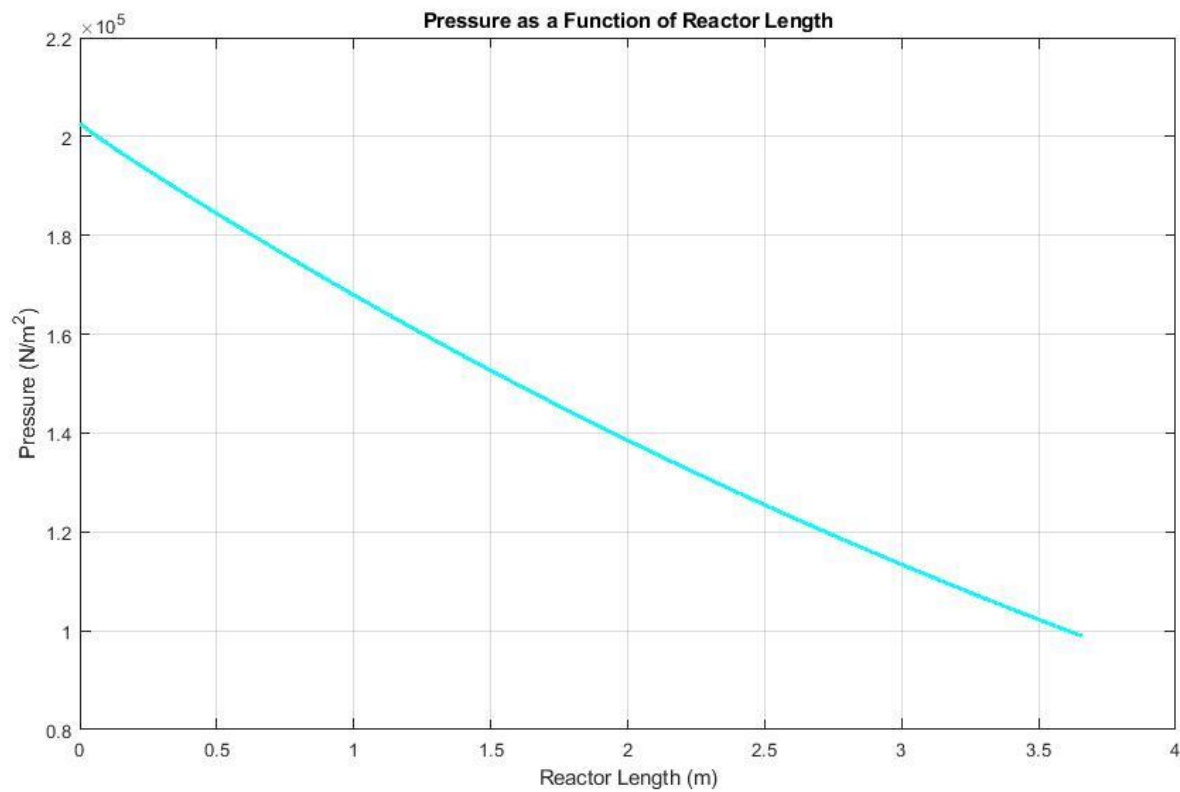
Conversion of Benzene:

$$X = \frac{F_{C_6H_{60}} - F_{C_6H_6}}{F_{C_6H_{60}}}$$

$$X = \frac{0.0041667 - 0.001873}{0.0041667} \times 100\%$$

$$X = 55.05\%$$





Part 2: Counter-Current Heat Exchanger

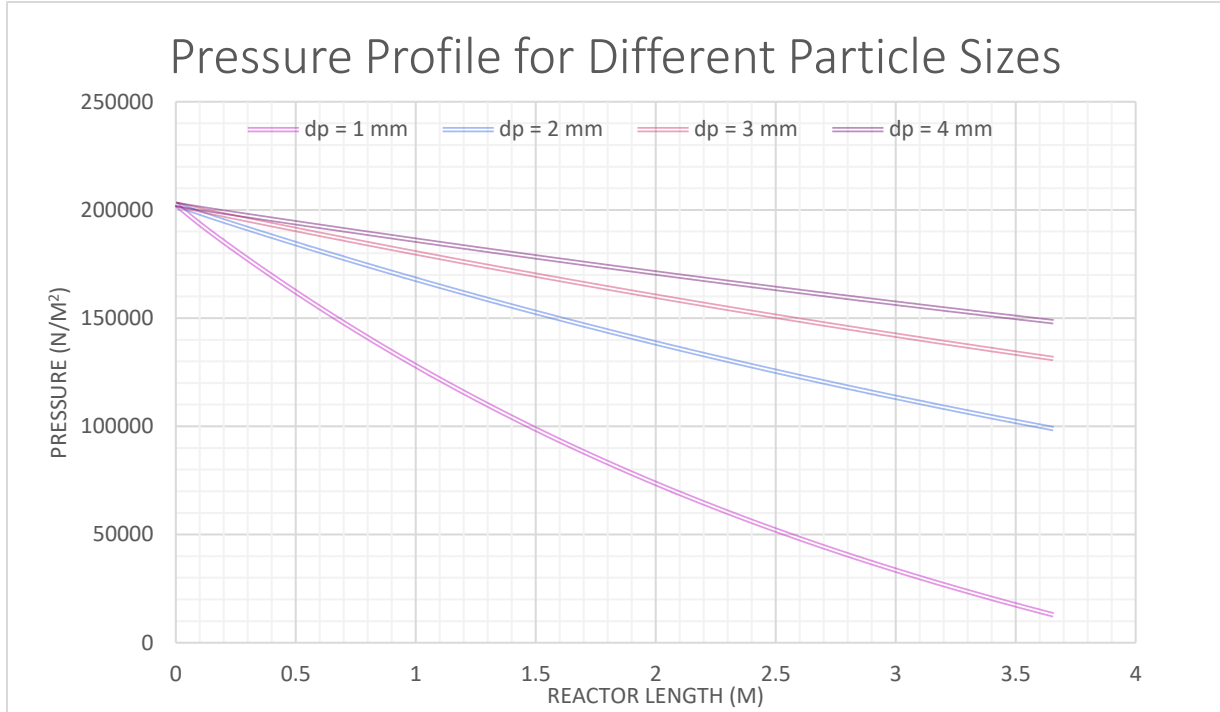
If there is a counter-current heat exchanger instead of co-current coolant the energy balance for the heat exchanger would become the following:

$$\frac{dT_a}{dz} = \frac{Ua(T - T_a)A_c}{\dot{m}_c c_{pc}} \rightarrow \frac{dT_a}{dz} = \frac{Ua(T_a - T)A_c}{\dot{m}_c c_{pc}} \quad (10)$$

In this case the reactants and coolant flow are in opposite direction, therefore the sign of the temperatures would switch. At the entrance of the reactor the reactants enter with a temperature of T_0 and the coolant exits at T_{af} . At the end of the reactor, the reactants and products exit at T_f and the coolant enters at T_{a0} .

Part 3: Graphs for varying particle sizes and inlet salt temperatures

a. The Pressure Profile for different particle sizes (1, 2, 3, and 4 mm)

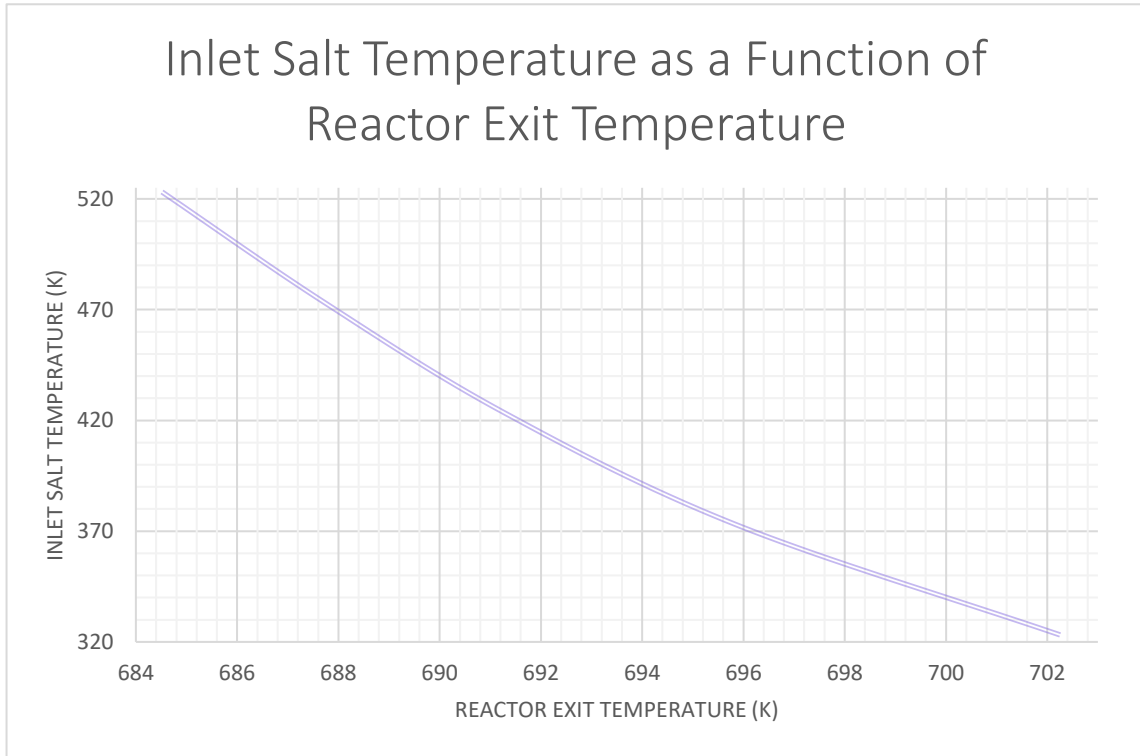


From these graphs it is clear, that as the particle size increases the pressure drop in the reactor decreases. Based on the Ergun equation, the results are as expected:

$$\frac{dP}{dz} = -\frac{G}{\rho_0 g_c d_p} \left[\frac{1 - \varepsilon}{\varepsilon^3} \right] \left[\frac{150(1 - \varepsilon)\mu}{d_p} + 1.75G \right] \frac{P_o}{P} \frac{T}{T_o} \frac{F_T}{F_{T_o}} \quad (8)$$

$as \ d_p \uparrow \ \frac{dP}{dz} \downarrow$

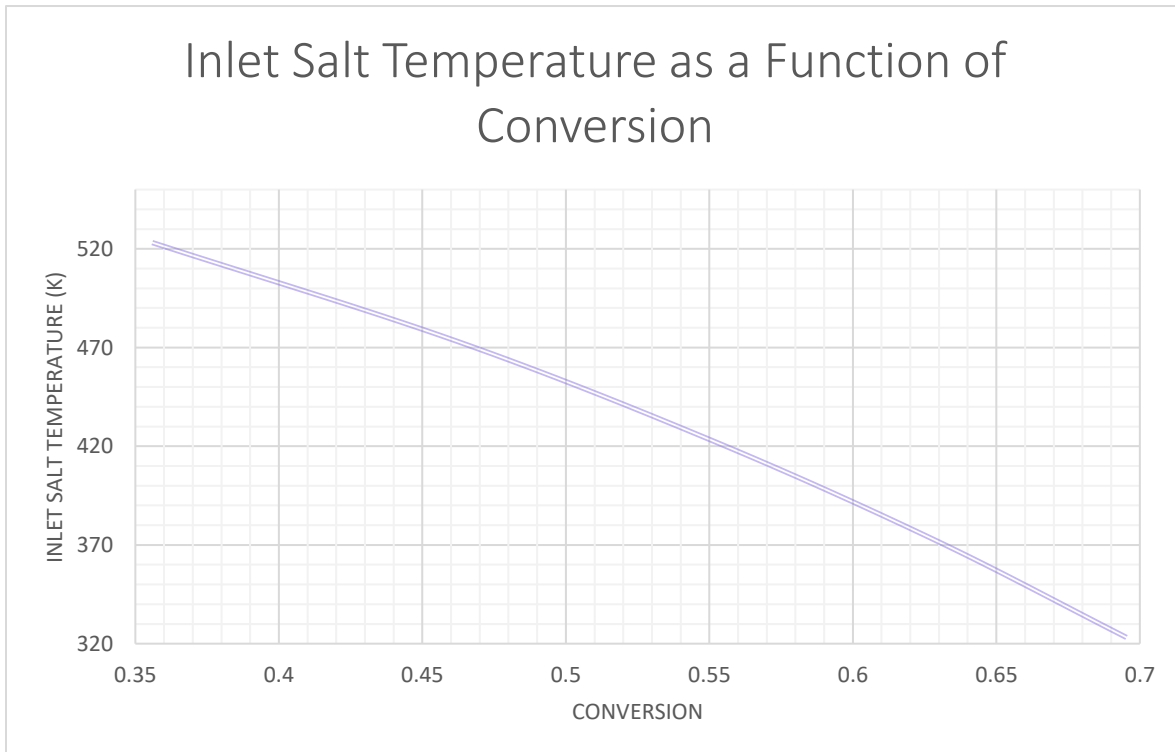
- b. Inlet salt temperature (50, 100, 150, 200, 250, 300 and 350 °C) as a function of the reactor exit temperature and final conversion (for 2 mm particle size).



The above graph shows the exit reactor temperature decreases as the inlet salt temperature increases. This is not correct, based on the energy balance equation for the coolant:

$$\frac{dT_a}{dz} = \frac{Ua(T - T_a)A_c}{\dot{m}_c c_{pc}} \quad (10)$$

Both the reactor temperature and salt temperature should increase and decrease simultaneously based on the above equation. This inaccurate graph may be due to an error in the code that was used in MATLAB.



The above graph displays that as inlet salt temperature increases conversion decreases. This is because if the cooling agent is increasing in temperature, it is no longer functioning properly, therefore, the conversion will be low. At a lower temperature, the cooling agent is operating correctly, and this will result in a higher conversion.

Part 4: Increase Conversion to 85-95%

In order to increase conversion to 85-95% many things can be done. Conversion increases when the pressure drop in the PBR decreases. If the diameter of the reactor is increased, then the pressure drop will decrease, causing a greater conversion. Increasing the particle size will also cause the same thing to occur, as demonstrated above. The pressure drop will decrease, and the conversion will increase. As stated above decreasing the inlet salt temperature also increases the conversion.

Part 5: Safety

Two significant parameters that are of concern when it comes to safety for this reaction are the inlet salt temperature and pressure drop. Increasing the inlet salt temperature will also cause an increase in the reactor exit temperature, as shown above. If the reactor temperature is increased significantly then it will go beyond the ignition point. This is very dangerous and can cause a runaway reaction. This reaction is thermally unstable, and the rate of heat generation exceeds the rate of cooling. A very high pressure drop is also very dangerous and should be avoided. One cause of high pressure drop is a high void fraction. Pressure drop will also increase if the particle diameter decreases. In order to account for these concerns the reactor must be design accordingly. If the inlet salt temperature is increased, it is necessary to make sure that the exit reactor temperature does not exceed the ignition point. For pressure drop the particle size should not be too small and the porosity or void fraction should not be too high.

CODE:

```
1 %Solve 10 ODEs for molar flow rates of 7 species, pressure drop,
2 %reactor temp, and coolant temp
3 function dFdz = pbr(z,F)
4
5 dFdz = zeros(10,1);
6 FC6H6 = F(1); %MOLAR FLOW RATE OF BENZENE
7 FC4H2O3 = F(2); %MOLAR FLOW RATE OF MALEIC ANHYDRIDE
8 FO2 = F(3); %MOLAR FLOW RATE OF OXYGEN
9 FCO2 = F(4); %MOLAR FLOW RATE OF CARBON DIOXIDE
10 FH2O = F(5); %MOLAR FLOW RATE OF WATER
11 FCO = F(6); %MOLAR FLOW RATE OF CARBON MONOXIDE
12 FN2 = F(7); %MOLAR FLOW RATE OF NITROGEN
13 P = F(8); %PRESSURE
14 T = F(9); %REACTOR TEMPERATURE
15 Ta = F(10); %SALT TEMPERATURE
16
17 %Given variables
18 k1 = 1.77*10^-5; %k1[=]k2 [=] mol/kg s (N/m^2)^.35 RATE CONSTANTS AT 400C
19 k2 = 2.66*10^-6;
20 k3 = 2.86*10^-8; %k3 [=] mol/kg s (N/m^2)
21 E12 = 83100; %All E [=] J/mol ACTIVATION ENERGY
22 E3 = 155000;
23 alpha = 0.35;
24 Do = 25*10^-3; %[=] m REACTOR OUTER DIAMETER
25 Di = 0.01865; %[=] m REACTOR INNER DIAMETER
26 Ac = pi*Di^2/4; %[=] m^2 CROSS-SECTIONAL AREA
27 rhob = 3.2*10^3; %[=] kg/m^3 BULK DENSITY
28 e = 0.45; %VOID FRACTION
29 dp = 2*10^-3; %m CATALYST DIAMETER
30 U = 120; %W/m^2 K OVERALL HEAT TRANSFER COEFFICIENT
31 delHR1 = -1236000; %All delH [=] J/mol HEAT OF REACTION
32 delHR23 = -2656000;
33 a = 4/Do; %[=] m^-1 HEAT-EXCHANGE AREA PER VOLUME OF REACTOR
34 mc = 19.52743605; %[=] mol/s COOLANT FLOW RATE
35 v0 = 0.00065317448; %[=] m^3/s INLET VOLUMETRIC FLOW RATE
36 rho0 = 1.230883923; %[=] kg/m^3
37 FT0 = 0.0277777778; %[=] mol/s INLET MOLAR FLOW RATE
38 FT = FC6H6+FC4H2O3+FO2+FCO2+FH2O+FCO+FN2; %[=] mol/s TOTAL MOLAR FLOW RATE
39 mu = 1.7894*10^-5*( (T/573.15)^1.5)*((573.15+110)/(T+110)); %[=] kg/m s VISCOSITY AS A FXN OF T
40 u = v0/Ac;
41 rho = rho0*(P/202650)*(573.15/T)*(FT0/FT);
42 G = rho*u;
43
44 %HEAT CAPACITIES FOR ALL SPECIES; SHOMATE EQUATION FROM ATHENA [=] J/mol K
45 cp1 = 4.184*(-8.101+113.3*10^-3*T-72.06*10^-6*T^2+17.03*10^-9*T^3); %C6H6
46 cp2 = 4.184*(-3.123+82.23*10^-3*T-52.17*10^-6*T^2+11.56*10^-9*T^3); %C4H2O3
47 cp3 = 4.184*(6.713-.0009*10^-3*T+4.17*10^-6*T^2-2.544*10^-9*T^3); %O2
48 cp4 = 4.184*(4.728+17.54*10^-3*T-13.38*10^-6*T^2+4.097*10^-9*T^3); %CO2
49 cp5 = 4.184*(8.0528-1.6761*10^-3*T+6.5196*10^-6*T^2-3.976*10^-9*T^3); %H2O
```

```

46 - cp2 = 4.184*(-3.123+82.23*10^-3*T-52.17*10^-6*T^2+11.56*10^-9*T^3); %C4H2O3
47 - cp3 = 4.184*(6.713-.0009*10^-3*T+4.17*10^-6*T^2-2.544*10^-9*T^3); %O2
48 - cp4 = 4.184*(4.728+17.54*10^-3*T-13.38*10^-6*T^2+4.097*10^-9*T^3); %CO2
49 - cp5 = 4.184*(8.0528-1.6761*10^-3*T+6.5196*10^-6*T^2-3.976*10^-9*T^3); %H2O
50 - cp6 = 4.184*(7.373-3.07*10^-3*T+6.662*10^-6*T^2-3.037*10^-9*T^3); %CO
51 - cp7 = 4.184*(7.44-3.24*10^-3*T+6.4*10^-6*T^2-2.79*10^-9*T^3); %N2
52 - cpc = 50+2*10^-3*Ta+1*10^-5*Ta^2-4*10^-9*Ta^3; %HEAT CAPACITY FOR MIXED SALT
53
54 - K2 = 0.003059462; %[=] (N/m^2)^-1 EQUILIBRIUM CONSTANT
55 - delcp2 = (3*cp4+6*cp5)-(7.5*cp3+cp1);
56 - K2T = K2*exp(((delHR1-673.15*delcp2)/8.314)*((1/673.15) - (1/T))+(delcp2/8.314)*log(T/673));
57
58 %RATE CONSTANTS AS A FXN OF TEMPERATURE
59 - k1T = k1*exp((E12/8.314)*((1/673.15)-(1/T)));
60 - k2T = k2*exp((E12/8.314)*((1/673.15)-(1/T)));
61 - k3T = k3*exp((E3/8.314)*((1/673.15)-(1/T)));
62
63 %DALTON'S LAW OF PARTIAL PRESSURES
64 - P1 = (FC6H6/FT)*P;
65 - P2 = (FC4H2O3/FT)*P;
66
67 %REACTION RATES
68 - r1 = (k1T*P1^alpha)/(1+K2T*P2)^2; %All [=] mol/kg s
69 - r2 = k2T*P1^alpha;
70 - r3 = (k3T*P2)/(1+K2T*P2)^2;
71
72 %SUMMATION OF EACH REACTION RATE TIMES THE CORRESPONDING HEAT OF REACTION
73 - sumrdelH = (r1*delHR1)+(r2*delHR23)+(r3*delHR23);
74 - sumFcp = FC6H6*cp1+FC4H2O3*cp2+FO2*cp3+FCO2*cp4+FH2O*cp5+FCO*cp6+FN2*cp7;
75
76 %MOLE BALANCES
77 - dFdz(1) = (-r1-r2)*Ac*rhob; %C6H6
78 - dFdz(2) = (r1-r3)*Ac*rhob; %C4H2O3
79 - dFdz(3) = (-4.5*r1-7.5*r2-2*r3)*Ac*rhob; %O2
80 - dFdz(4) = (2*r1+6*r2+2*r3)*Ac*rhob; %CO2
81 - dFdz(5) = (2*r1+3*r2+r3)*Ac*rhob; %H2O
82 - dFdz(6) = 2*r3*Ac*rhob; %CO
83 - dFdz(7) = 0; %N2
84
85 %ERGUN EQUATION
86 - dFdz(8) = -((G*(1-e))/(rho0*dp*e^3))*(((150*(1-e)*mu)/dp)+1.75*G)*(202650/P)*(T/573.15)*(FT/FT0);
87
88 %ENERGY BALANCES
89 - dFdz(9) = ((sumrdelH+((U*a*(T-Ta))/rhob))/sumFcp)*Ac*rhob; %REACTOR TEMPERATURE
90 - dFdz(10) = (U*a*(T-Ta)*Ac)/mc/cpc; %SALT TEMPERATURE
91
92 - dFdz == dFdz(:)
93 - end
94

```

SCRIPT:

```
1 - clear all
2 - close all
3 - length = [0 3.6576];
4 - IC = [0.004166667, 0, 0.011805556, 0, 0, 0, 0.011805556, 202650, 573.15, 423.15];
5 - [z,F] = ode45(@pbr,length,IC); %SOLVE THE 10 ODEs, 7 FLOW RATES, 1 PRESSURE DROP, 1 REACTOR-TEMP, AND 1 SALT-TEMP
6
7 %PLOT OF MOLAR FLOW RATES AS FXN OF Z
8 - set(0,'DefaultAxesColorOrder',cool(7))
9 - plot(z,F(:,1),'linewidth',2);
10 - hold all
11 - plot(z,F(:,2),'linewidth',2);
12 - hold all
13 - plot(z,F(:,3),'linewidth',2);
14 - hold all
15 - plot(z,F(:,4),'linewidth',2);
16 - hold all
17 - plot(z,F(:,5),'linewidth',2);
18 - hold all
19 - plot(z,F(:,6),'linewidth',2);
20 - hold all
21 - plot(z,F(:,7),'linewidth',2);
22 - hold all
23 - title('Molar Flow Rates of Each Component as a Function of Reactor Length')
24 - xlabel('Reactor Length (m)')
25 - ylabel('Flow Rate (mol/s)')
26 - legend('F_{C_6H_6}','F_{C_4H_2O_3}','F_{O_2}','F_{CO_2}','F_{H_2O}','F_{CO}','F_{N_2}')
27 - set(gcf, 'Position', [100, 100, 1000, 600])
28 - grid on
29 - figure;
30
31 %PLOT OF PRESSURE AS A FUNCTION OF Z
32 - plot(z,F(:,8),'linewidth',2,'color','m');
33 - title('Pressure as a Function of Reactor Length')
34 - xlabel('Reactor Length (m)')
35 - ylabel('Pressure (N/m^2)')
36 - set(gcf, 'Position', [100, 100, 1000, 600])
37 - grid on
38 - figure;
39
40 %PLOT OF REACTOR-TEMP AND SALT-TEMP AS A FUNCTION OF Z
41 - plot(z,F(:,9),'linewidth',2);
42 - hold all
43 - plot(z,F(:,10),'linewidth',2);
44 - title('Temperature as a Function of Reactor Length')
45 - xlabel('Reactor Length (m)')
46 - ylabel('Temperature (K)')
47 - legend('Reactor-Temperature','Salt-Temperature')
48 - set(gcf, 'Position', [100, 100, 1000, 600])
49 - grid on
```